HEALTH HACKATHON 3.0 / VIT BHOPAL

VITALINK

The offline maternal early-warning network

Safer screening. Faster referrals.

Connected care - even without internet.



PROJECT CONCEPT • PITCH PRESENTATION

THE PROBLEM

The risk is not just detection. It is the handoff.

A concerning reading can become a delayed referral when systems and connectivity fail.

01 INCONSISTENT CAPTURE

Vitals and symptoms may be recorded without consistent quality checks.

02 LATE ESCALATION

Severe readings need a clear and immediate clinical workflow.

03 BROKEN REFERRAL

A saved record is not proof that the hospital received it.

Target users: ASHA workers • ANMs • Primary health centres • Regional clinicians

THE SOLUTION

One workflow. From village to clinician.

A clinically constrained product, not a camera-only diagnostic claim.

1

MEASURE

Validated BP and structured symptoms

2

CHECK

Quality checks and safety rules

3

ASSESS

Optional calibrated edge AI

4

REFER

Durable offline referral

5

CONFIRM

Clinician acknowledgment

WORKS WITHOUT INTERNET

CLINICAL SAFETY

Rules first. AI never overrides an emergency.

A severe-range BP finding must trigger urgent clinical assessment regardless of model score.

DETERMINISTIC SAFETY

Severe-range pregnancy BP: systolic ≥160 or diastolic ≥110 mmHg. Escalate under local clinical protocol.

UNCERTAINTY-AWARE AI

For non-emergency cases, calibrated prediction may assist prioritization. Uncertain results abstain.

No diagnosis claims • No fabricated accuracy • Human clinical oversight required

TECHNICAL DIFFERENTIATION

A small model with a big safety envelope.

Clinical rule engine + calibrated risk + selective prediction + measured device performance.

MODEL

Logistic baseline; XGBoost candidate; ONNX Runtime Mobile.

UNCERTAINTY

Calibration, conformal prediction, and explicit abstention.

EXPLAINABILITY

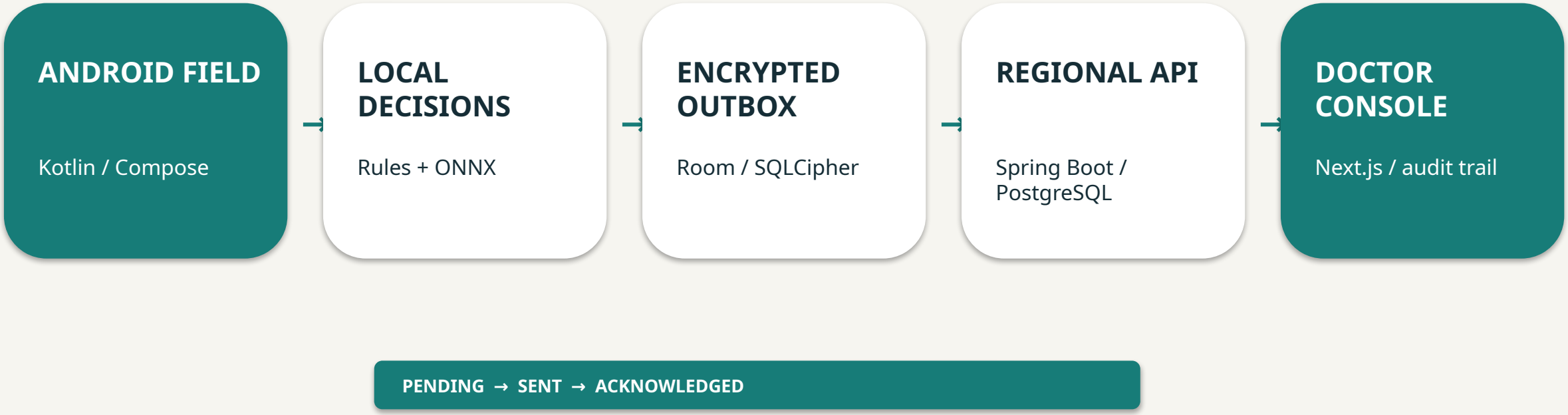
Feature contributions with honest confidence and model limitations.

Key evaluation: sensitivity, specificity, calibration, false negatives, and site shift.

ARCHITECTURE

Edge autonomy. Reliable synchronization.

Local screening continues during a network outage; server acknowledgment is a separate event.



LIVE DEMO

180 seconds. One complete case.

Airplane mode first. A synthetic severe reading. Then a verified hospital acknowledgment.

00:00

OFFLINE

Turn on airplane mode

00:40

ASSESS

Enter synthetic BP 166/112

01:20

REFER

Save a pending referral

02:00

SYNC

Restore connectivity

02:30

ACK

Doctor confirms receipt

Demo data are synthetic; a rules-based trigger is not a validated diagnosis.

HEALTH ECONOMICS

Measure the workflow, not imaginary savings.

Equipment reuse lowers per-screening allocation; clinical and operational outcomes need field evidence.

ILLUSTRATIVE UNIT ECONOMICS

₹7.78

per screening: ₹10,000 / (150 × 24) + ₹5

WHAT TO ACTUALLY MEASURE

Referral acknowledgment time, documentation time, duplicate records, missed handoffs, and staff burden.

Scenario only. Excludes labor, training, maintenance, and clinical follow-up.

EVIDENCE

Prove reliability before making clinical claims.

Engineering acceptance tests are achievable during the hackathon; patient outcomes require future studies.

OFFLINE

100% supported scenarios

LATENCY

p95 < 300 ms target

REFERRALS

No lost committed records

DUPLICATES

Zero in retry tests

RULES

All reviewed fixtures pass

PRIVACY

No PII in app logs

ROADMAP

Five milestones. One deployable demonstration.

The submission is a verified prototype with a documented clinical boundary.

01 CLINICAL FOUNDATION

02 OFFLINE REFERRALS

03 REGIONAL WORKSPACE

04 EDGE AI + VALIDATION

05 RELEASE + SUBMISSION

VITALINK: recognize danger, act locally, and close the referral loop.